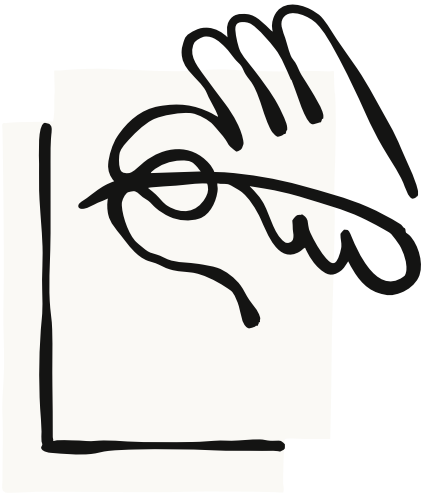
Announcements

How Claude’s text watermark works

Aug 14, 2026



Future Claude models will generate text that contains a watermark. This is a way of determining the likelihood that Claude was involved in writing the text, and we, along with several other major AI providers, are implementing this change to comply with the EU AI Act.

In this article, we share answers to some of the questions we’ve received about how our chosen watermarking method works, whether it affects Claude’s outputs, and why we’re making this change. To summarize:

- We use a method of watermarking that does not have any practical impact on the quality or content of Claude’s outputs;
- The difference between watermarked and un-watermarked text will not be distinguishable to readers;

- Nothing is added to the text and there are no hidden characters;
- Watermarking doesn't require extra tokens, and will not be more expensive;
- Watermarking carries no identifying information and can't be traced to a specific person, organization, or chat;
- Watermarking won't be specific to Claude. As of August 2, the EU requires AI providers serving its market to mark AI-generated content. Other major model developers have signed the same Code of Practice and will be implementing their own watermarks.

What is watermarking?

Large language models like Claude work by generating one word at a time. Each time the model decides on the next word, it chooses among a list of potential candidates, ultimately selecting the most sensible or likely based on the preceding text. Take the sentence "The weather today was cold and…". The next word is very unlikely to be "sugary." But it is quite likely to be "overcast" or "grey." Under most circumstances, it doesn't matter much to the reader which of these latter two words the model ultimately chooses—the meaning of the sentence is largely the same either way. In cases like this, the choice is settled by a random number.

Watermarking uses low-stakes choices like these—which occur many times over a piece of generated text—to leave a pattern in Claude's responses. That pattern is undetectable to the reader, but *is* detectable to anyone who has a key that encodes it. When watermarking is used, choices are still made at random, but the *source* of the randomness is different. Instead of using an arbitrary random number generator to pick the next word, watermarking uses the key and a few words that come before to settle what word the model should pick. That is, the words that Claude picks are still random, but now, one can check the sequence of words and see if it's consistent with the choices Claude would make if it was using the key. If it is, one can assign a probability that the text was generated by Claude.

Importantly, it isn't that the model will now always be biased toward overcast or grey. Just as with non-watermarked text, overcast might be selected in one sentence, grey in the next, depending on the words that came before. And it's not the case that the watermarking method pushes Claude to choose a word it wouldn't have considered anyway (for instance, it wouldn't make Claude pick a word like "nubilous"—an obscure[1] synonym for overcast or grey that Claude almost certainly wouldn't use under normal circumstances).

How does watermarking affect Claude's outputs?

Watermarking does not impact the quality of Claude's output. To a reader, a watermarked response is indistinguishable from an unwatermarked one (in this way, AI watermarks differ substantially from their namesakes on banknotes, other physical objects, and some digital documents, which *are* visible to the naked eye).

In internal testing, we've seen no impact of watermarking on the content, level of creativity, or readability of Claude's text. In the SynthID-Text paper, which introduced the technique we use, Google DeepMind tested this impact by serving a model that used watermarking to a portion of their Gemini traffic and comparing thumbs-up and thumbs-down ratings. They found no statistically significant differences from the unwatermarked model. And in a controlled study, human raters comparing watermarked and unwatermarked answers side-by-side saw no difference in quality.

A useful analogy is to imagine you're playing a game like Monopoly. On each turn, each player moves a random number of spaces around the board according to the roll of a die. Suppose that, instead of rolling the die to get this randomness, we decided to use a book of the digits of pi.[2] We start from a randomly-chosen digit (say, the 1,012,845th after the decimal place, which happens to be a 6), and from that point on each player simply uses the next digit in the sequence as their next "roll."

For all intents and purposes, the moves are still random: it makes no difference to the players—or to the outcome of the game—whether the randomness comes from pi or from dice rolls each time. But if we could see the sequence of all the moves after the game (and we knew the value of pi), we could work out whether this was a game that likely used pi to determine its moves. The game that used pi is, in a sense, "watermarked."

It's the same for Claude-generated text. Watermarking doesn't change the meaning or experience for the person reading it, but if you wanted to check after the fact whether the text was likely generated by Claude, the watermark allows you to do so.

Which specific method of watermarking do you use?

Claude's text watermark is a version of the SynthID-Text approach published by Google DeepMind in a *Nature* paper in 2024. It belongs to a family of approaches that go back to a proposal by Scott Aaronson in 2022, all of which share the same

design principle that we described above—the watermark only changes the source of the randomness used to pick among words.

There are limitations to the effectiveness of watermarking. Using our key, one can only answer the question "What is the likelihood this was partly written by Claude?" It doesn't confirm whether the text was human-written, and it can't tell whether the text was written by a different AI (even if that other AI uses watermarking, it would have a different key; it might also use a different watermarking method altogether). Detecting a watermark also doesn't work well on small samples, where there are fewer word choices and thus less information to go on. As a passage increases in length, confidence about Claude's involvement increases too.

Watermarking is sparser on factual passages where there are fewer choices that can be made without decreasing the accuracy of the text. For example, take the sentence "Isaac Newton's most famous work was called *Principia*…". It really matters whether the next word is "*Mathematica*" (it's the only right answer), so the watermark would have nothing to act on. The same is true for proofreading. If you hand Claude a piece of writing and ask it to edit only the grammar and punctuation and nothing else, the watermark can only live in the handful of corrections, which might be too few to register.

What about cases where Claude has proofread or edited human text?

The watermark only applies to words Claude chooses. When Claude proofreads text written by a person, what it gives back has generally only been lightly edited; because nearly all the words are the person's, there's very little (if anything) for the watermark to attach to. Depending on the length of the text and how heavily Claude has edited it, those changes might not be enough to make Claude's involvement detectable. The more Claude writes, the more decisions it has to make, and the more space there is for a watermark.

What about code?

As we noted above, AI watermarking takes advantage of decisions where either choice of a word would be equally good. Where an *exact* output is required—where there isn't a choice, and something would be factually wrong or a piece of code would break if a different term was chosen—the watermark isn't applied.

For example, once the model has written "2 + 2 =", there is a very clear best choice for the next token (if the model is completing the sum, there isn't an answer that's

equally as good as "4"; if it's talking about George Orwell's *Nineteen Eighty-Four*, there isn't an answer that's equally as good as "5"). The "nudge" of the watermark wouldn't be applied here. For the same reason, code—which in very many cases has to be exact—has generally less watermarking than some other forms of text.

Having said that, in areas where there *is* an arbitrary choice between particular words or terms within the code, the watermark can be used, such as comments within code. But by definition, it will have a negligible effect on the actual code produced.

What does this mean for users?

Does this slow the model down, or make it more expensive?

No. Watermarking has a negligible impact on the speed of models, and because it produces no extra tokens, the model is the same price to serve and use.

Can a watermark be traced back to me or my organization?

No. The watermarking applies to Claude and its outputs. It doesn't identify anything to do with individual users. There's nothing in the watermark, or its key, that would allow anyone to recover any information about the user, their organization, or their chats with Claude.

Why are you watermarking Claude's outputs?

We're implementing watermarking to comply with the EU AI Act. Anthropic, along with several other major AI model providers and around 190 total signatories, signed the EU Code of Practice on Transparency of AI-Generated Content in July 2026. This requires AI system providers to use methods of "marking" AI-generated text. We're applying watermarking globally at launch because we don't yet have a durable way to scope it by region. However, we will continue to evaluate different approaches, and will share updates when we have them.

Other questions

How do I check if a piece of text was written by Claude?

We are releasing a detection API in private preview. It is currently available to eligible organizations as required under EU law (such as regulators, law enforcement, media, fact-checkers, independent researchers, educational

organizations, and EU civil society groups). It is also available for enterprises who are similarly obligated to verify watermarking for their own compliance with the Act. We plan to expand access to the detection API over time. You can register interest in access here.

What about images and other files?

When Claude produces a file of a supported type (such as a .png, .jpg, or .svg), it will attach a content credential in the form of a small, cryptographically signed note in the file's metadata, saying that the file was made or processed with Claude. This is an open industry standard called C2PA—the same used by camera manufacturers and in photo-editing software to record where an image came from. Any C2PA-aware tool can read it; we'll be providing our own where you can drop a file and check.

This metadata label is very different from a watermark. Nothing in the file changes—it is not embedded or hidden. As with text, the credential only says Claude was involved in producing the file; it doesn't include any identifying information.

Can't someone just edit the text to get around the watermarking?

To some extent, yes. Light editing probably won't remove the watermark completely; a complete rewrite where every word is replaced will. In the latter case, of course, it's arguable whether the text can any longer be described as AI-generated.

What does a watermark actually prove?

A watermark can only determine that Claude was likely involved with the content at some point. It cannot distinguish "Claude wrote this" from "Claude heavily edited this."

Do watermarks apply to translations?

Yes. A translation produced by Claude carries a watermark, because in this case every word is chosen by Claude.

What about older Claude models?

The EU law includes a transition period for Anthropic models launched before August 2, 2026, and we're working to add watermarking for those models as well. This will be rolled out over the coming months.

How does this differ from AI detection software, like Pangram?

AI detection software uses a different method, because the companies that provide it don't have our key. Among other things, those services look at aspects of the text like the subtle (and not-so-subtle) "tells" that often appear in AI's phrasing. For example, AI models appear to be fond of the construction "this isn't [X], it's [Y]", and use the word "quietly" a lot more than you might expect. Picking up on these patterns is fundamentally different from checking for a watermark.

Does this change who owns a given output, or who is legally responsible for it?

No. A watermark only helps test whether Claude might have produced or processed the content. It doesn't say anything about ownership or authorship, and doesn't change a user's rights under our terms. We only apply the watermark when Claude was involved in processing the content or file.

Updated Sep 1, 2026: Provided up to date information on the watermarking detection API.

Footnotes

1. Or, you might say, nubilous—which is also a synonym for "obscure." ↩
2. Pi is technically predictable, but any run of digits from somewhere in the middle of pi is indistinguishable from a run of rolls of a ten-sided die. Also, set aside the fact that the dice in Monopoly go from 1 to 6 whereas a digit of pi can be from 0 to 9; the analogy isn't perfect. ↩

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